Analytical Chemistry 1 (125101)

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Framework of this course

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❖ Section I Chemical Measurements

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Section I

Chemical Measurement

1.1 Basic knowledge

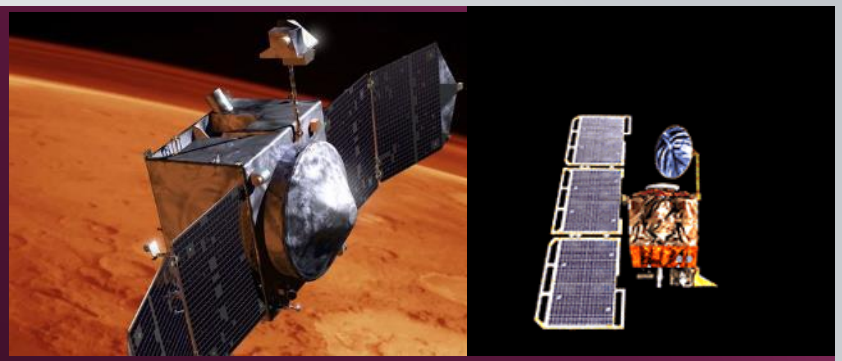
Overview

1.1-1 Common prefixes and SI units

1.1-2 Chemical concentration

1.1-3 Preparing solutions

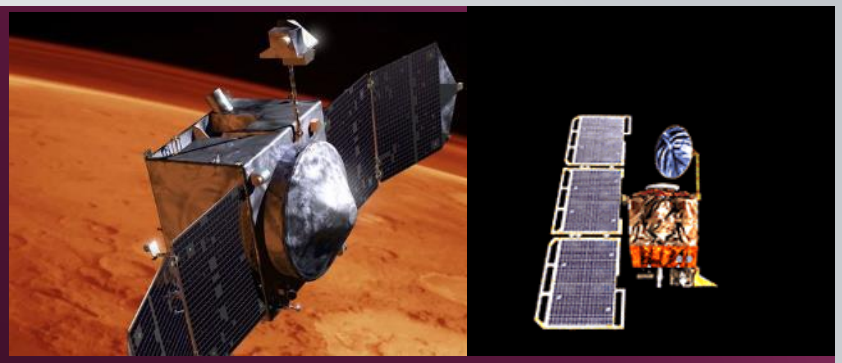
1.1-4 Stoichiometry calculations for limiting reagents



Oops!

In 1999, the $125 million Mars climate Orbiter spacecraft was lost when it entered the Martian atmosphere 100 km lower than planned. This navigation error would have been avoided if people had written their units of measurement. Engineers who built the spacecraft calculated thrust in the English unit, pounds of force. Jet propulsion laboratory engineers thought they were receiving the information in the metric unit, Newtons. Nobody caught this error.

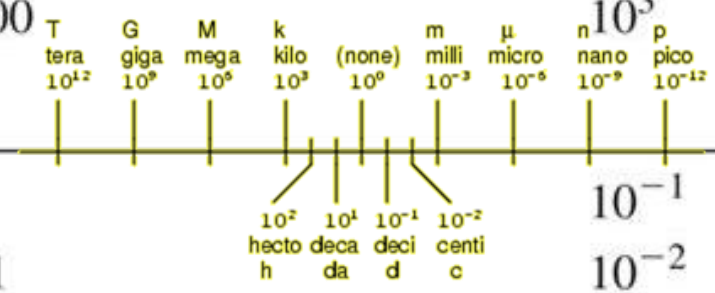
Oops!



This story tells us it is veeeeery necessary to carry units along with the numbers when you do any calculation!

Common Prefixes with SI Units

Prefix[a]	Symbol	Meaning	Numerical Value	Scientific Notation
Prefixes That Increase the Size of the Unit				
tera	T	trillion	1 000 000 000 000	10^{12}
giga	G	billion	1 000 000 000	10^{9}
mega	M	million	1 000 000	10^{6}
kilo	k	thousand	1 000	10^{3}
Prefixes That Decrease the Size of the Unit				
deci	d	tenth	0.1	10^{-1}
centi	c	hundredth	0.01	10^{-2}
milli	m	thousandth	0.001	10^{-3}
micro	μ	millionth	0.000 001	10^{-6}
nano	n	billionth	0.000 000 001	10^{-9}
pico	p	trillionth	0.000 000 000 001	10^{-12}
femto	f	quadrillionth	0.000 000 000 000 001	10^{-15}



[a] The prefixes most often used by chemists are in boldface type.

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1.1-1 SI Units

SI base units include the following:

- meter (m) → Length
- kilogram (kg) → Mass
- second (s) → Time
- ampere (A) → Electric current
- kelvin (K) → Temperature
- mole (mol) → Amount of substance

SI — International system (i.e. metric system)

1.1-1 SI Units

SI base units

Symbol	Name	Quantity
s	second	time
m	metre	length
kg	kilogram	mass
A	ampere	electric current
K	kelvin	thermodynamic temperature
mol	mole	amount of substance
cd	candela	luminous intensity

SI defining constants

Symbol	Name	Exact value
$\Delta\nu_{\mathrm{Cs}}$	hyperfine transition frequency of Cs	9 192 631 770 Hz
c	speed of light	299 792 458 m/s
h	Planck constant	$6.626\,070\,15 \times 10^{-34}$ J·s
e	elementary charge	$1.602\,176\,634 \times 10^{-19}$ C
k	Boltzmann constant	$1.380\,649 \times 10^{-23}$ J/K
N_{A}	Avogadro constant	$6.022\,140\,76 \times 10^{23}$ mol^{-1}
K_{cd}	luminous efficacy of 540 THz radiation	683 lm/W

Derived SI Units

Other quantities can be derived in terms of base SI units.

- force (newton, N), kg·m / s^2
- pressure (pascal, Pa), N / m^2
- energy (joule, J), N·m

Volume: Liter (L)

1 L = 1 dm × 1 dm × 1 dm = 0.1 m×0.1 m×0.1 m = 0.001 m^3

1 mL = 0.001 L = 1 cm ×1 cm×1 cm = 1 cm^3

1 J =1 N·m = 1 N·m^3 /m^2 = 1 Pa·m^3 = 1000 Pa·L

Temperature 0 °C = 273.15 K, K = 273.15 + °C

Using Prefixes in Calculations

- In calculations, units should be carried along with the numbers.
- Prefixes such as kilo- and milli- are used to denote multiples of units.
- Prefixes can be used to simplify conversions between units.

Example:

Express 19.3 mPa in terms of atm

19.3 mPa can be written as 19.3×10^{-3} Pa

Pressure **1 atm = 101325 Pa = 760 mm Hg = 760 torr**

1 bar = 10^5 Pa

$$19.3 \times 10^{-3}\ \text{Pa} \times \frac{1.00\ \text{atm}}{101325\ \text{Pa}} = 1.90 \times 10^{-7}\ \text{atm}$$

1.1-2 Concentrations

- A great deal of chemical analyses take place in solution;
- It is imperative to know the concentration of solution;
- Concentration is an intrinsic characteristic of a solution;
- Solutions are homogeneous mixtures. So, it doesn't change when the solution is divided into several portions.
  (in contrast to weight and volume)

1.1-2 Concentrations

- **Molarity** (moles of solute per liter of solution), M
- **Molality** (moles of solute per kilogram of solvent), m
- **Formal concentration** (formula units per liter), F
- **Percent composition** (w/w, v/v %) w/v, g/mL
- **Parts per million,** ppm
- **Parts per billion,** ppb

% Composition (w/w or v/v)

$$\text{Weight percent} = \frac{\text{mass of solute}}{\text{mass of total solution or mixture}} \times 100$$

$$\text{Volume percent} = \frac{\text{volume of solute}}{\text{volume of total solution}} \times 100$$

95% (w/w) ethanol contains 95 g of ethanol for every 100 g of solution.

ppm, Parts per Million

$$\text{ppm} = \frac{\text{mass of substance}}{\text{mass of sample}} \times 10^6$$

$$\text{ppb} = \frac{\text{mass of substance}}{\text{mass of sample}} \times 10^9$$

- **Shortcut:** ppm is also mg/L or µg/mL if the density of the solution is 1.0 g/mL.
- An aqueous solution that is 1 000 ppm in Cu contains 1 000 mg Cu per liter of solution.
- It also contains 1000 µg per mL of solution.

ppm, Parts per Million

$$\text{ppm} = \frac{\text{mass of substance}}{\text{mass of sample}} \times 10^6$$

$$\text{ppb} = \frac{\text{mass of substance}}{\text{mass of sample}} \times 10^9$$

For aqueous solution

ppm = mg/L = μg/mL

1 g H_2O equates to 1 mL H_2O

$\rho(H_2O) = 1.0$ g/mL

ppb = μg/L = ng/mL

ppm means parts per million (0.001‰)

% means part per hecto (a factor of one hundred)

ppm is dimensionless, if the solution density is 1 g/mL

The concentration: parts per million × 1 g/mL

1 ppm=1 millionth g/mL=1 μg/mL

EXAMPLE Converting Parts per Billion into Molarity

Normal alkanes are hydrocarbons with the formula C_nH_{2n+2}. Plants selectively synthesize alkanes with an odd number of carbon atoms. The concentration of $C_{29}H_{60}$ in summer rainwater collected in Hannover, Germany, is 34 ppb. Find the molarity of $C_{29}H_{60}$ and express the answer with a prefix

Solution A concentration of 34 ppb means there are 34 ng of $C_{29}H_{60}$ per gram of rainwater, which is nearly the same as 34 ng/mL because the density of rainwater is close to 1.00 g/mL. To find the molarity, we need to know how many grams of $C_{29}H_{60}$ are contained in a liter. Multiplying nanograms and milliliters by 1 000 gives 34 μg of $C_{29}H_{60}$ per liter of rainwater:

$$\frac{34\ \cancel{\text{ng}}\ C_{29}H_{60}}{\cancel{\text{mL}}}\left(\frac{1\ 000\ \cancel{\text{mL}}/\text{L}}{1\ 000\ \cancel{\text{ng}}/\mu\text{g}}\right) = \frac{34\ \mu\text{g}\ C_{29}H_{60}}{\text{L}}$$

The molecular mass of $C_{29}H_{60}$ is $29 \times 12.011 + 60 \times 1.008 = 408.8$ g/mol, so the molarity is

$$\text{Molarity of } C_{29}H_{60} \text{ in rainwater} = \frac{34 \times 10^{-6}\ \cancel{\text{g}}/\text{L}}{408.8\ \cancel{\text{g}}/\text{mol}} = 8.3 \times 10^{-8}\ \text{M}$$

An appropriate prefix from Table 1-3 would be nano (n), which is a multiple of 10^{-9}:

$$8.3 \times 10^{-8}\ \cancel{\text{M}}\left(\frac{1\ \text{nM}}{10^{-9}\ \cancel{\text{M}}}\right) = 83\ \text{nM}$$

TEST YOURSELF: How many ppm of $C_{29}H_{60}$ are in 23 μM $C_{29}H_{60}$? (9.4 ppm)

1.1-3 Making Solutions

- To calculate quantities of reagents needed to prepare solutions, use the relation

$$\underbrace{(M_{conc})(V_{conc})}_{\text{Moles taken from concentrated solution}} = \underbrace{(M_{dil})(V_{dil})}_{\text{Moles placed in dilute solution}}$$

Dilution formula

- Equates moles of reagent removed from a stock solution to moles delivered into a new solution.

Making Solutions

- How many **milliliters** of 10 000 ppm Cu^{2+} needed to make 500 mL of a solution that is 1 000.0 ppm in Cu^{2+}

- **moles** of Cu^{2+}

$(M_{conc})(V_{conc}) = (M_{dil})(V_{dil})$

$(10\ 000\ ppm)(V_{conc}) = (1\ 000\ ppm)(500\ mL)$

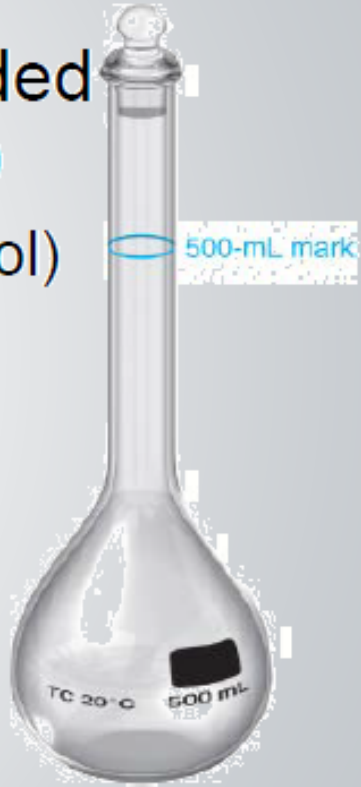
$V_{conc} = (1\ 000\ ppm)(500\ mL) / (10\ 000\ ppm)$

$V_{conc} =$ 50 mL

Place 50 mL of 10 000 ppm Cu^{2+} stock solution in the 500 mL Volumetric flask and fill H_2O to the 500 mL mark of the flask.

Making Solutions

- How many grams of $CuCl_2 \cdot 5H_2O$ are needed to make 500 mL of a solution that is 1 000 ppm in Cu? (Mw of $CuCl_2 \cdot 5H_2O$ is 224.53 g/mol)



- First calculate the grams of Cu required
  - (500-mL) × (1000 µg/mL) = 5.0×10^5 µg Cu
  - 5.0×10^5 µg Cu = $(5.0 \times 10^5)(10^{-6})$g Cu
  - 5.0×10^{-1}g Cu

$$\left(5.0 \text{ x } 10^{-1} \text{ g Cu}\right) \times \left(\frac{224.53 \text{ g } CuCl_2 \cdot 5H_2O \text{ / mol}}{63.546 \text{ g Cu /mol}}\right) = 1.76_7 \text{ g } CuCl_2 \cdot 5H_2O$$

1.1-4 Stoichiometry

- Use stoichiometry relationships to calculate required masses or volumes of reagents for chemical reactions.
- From the mass of product of a reaction, you should be able to compute how much reactant was consumed.

Stoichiometry

- Iron from dietary supplement tablet can be measured by gravimetric method

- Dissolving tablet and converting dissolved iron into solid Fe_2O_3

- Mass of Fe_2O_3 tells us the mass of iron in the original tablet

Stoichiometry

Dissolve ⇨ Oxidize ⇨ Precipitate ⇨ Sinter ⇨ Weigh

①$2Fe^{2+} + H_2O_2 + 2H^+ \rightarrow 2Fe^{3+} + 2H_2O$

②$Fe^{3+} + 3OH^- + (x-1)H_2O \rightarrow FeOOH \cdot xH_2(\downarrow)$

③$FeOOH \cdot xH_2O(s) \rightarrow Fe_2O_3(s)$

Each tablet provides ~ 15 mg of iron, how many tablet we need to obtain 0.25 g of Fe_2O_3 product?

0.25 g of Fe_2O_3 (MW= 160 g/mol) product contains

$=0.25\ g \times \frac{56\times2\ g/mol}{160\ g/mol} = 0.175\ g\ Fe$

Each Tablet provides about 15 mg of Fe

We need how many pieces of tablet

=0.175g/0.015g = 11.67 pieces

As a result, 12 pieces of tablet need to be dissolved to obtain 0.25 g of Fe_2O_3 product.

Limiting Reagent

- The **limiting reagent** in a chemical reaction is the one that is consumed first.
- Once the limiting reagent is gone, the reaction ceases.

Example: For the reaction: $A + 2B \rightarrow P$

0.751 moles of A are mixed with 1.43 moles of B.

- What is the limiting reagent and how much excess reagent remains unreacted?

Limiting Reagent

Solution:

	A	+	2B	→	P
I.	0.751		1.43		0
F.	?		?		

If A is the L.R., how much B is required to completely consume A?

$$0.751 \text{ mol A} \times \left(\frac{2 \text{ mol B}}{1 \text{ mol A}} \right) = 1.502 \text{ mol B required}$$

However, we only have 1.43 mol B, so B must be the L.R.

Limiting Reagent

Solution:

	A	+	2B	→	P
Initial	0.751		1.43		0
Final	?		0		

If B is the L.R., how much is required to completely consume A?

$$0.1.43 \text{ mol B} \times \left(\frac{1 \text{ mol A}}{2 \text{ mol B}} \right) = \mathbf{0.715\ mol\ A}$$ react with B.

So, 0.751 mol A – 0.715 mol A = 0.036 mol A remain after the reaction is complete. All of B is consumed.

EXAMPLE Limiting Reagent

Reaction 1-9 requires one mole of oxalate for each mole of calcium.

$$Ca^{2+} + \underset{\text{Oxalate}}{C_2O_4^{2-}} \rightarrow \underset{\text{Calcium oxalate}}{Ca(C_2O_4) \cdot H_2O(s)} \qquad (1\text{-}9)$$

If you mix 1.00 g of $CaCl_2$ (FM 110.98) with 1.15 g of $Na_2C_2O_4$ (FM 134.00) in water, which is the limiting reagent? What fraction of the nonlimiting reagent is left over?

Solution The available moles of each reagent are

$$\frac{1.00 \text{ g } CaCl_2}{110.98 \text{ g/mol}} = 9.01 \text{ mmol } Ca^{2+} \qquad \frac{1.15 \text{ g } Na_2C_2O_4}{134.00 \text{ g/mol}} = 8.58 \text{ mmol } C_2O_4^{2-}$$

The reaction requires 1 mol Ca^{2+} for 1 mol $C_2O_4^{2-}$, so oxalate will be used up first. The Ca^{2+} remaining is $9.01 - 8.58 = 0.43$ mmol. The fraction of unreacted Ca^{2+} is (0.43 mmol/9.01 mmol) = 4.8%

TEST YOURSELF The reaction $5H_2C_2O_4 + 2MnO_4^- + 6H^+ \rightarrow 10CO_2 + 2Mn^{2+} + 8H_2O$ requires 5 mol $H_2C_2O_4$ for 2 mol MnO_4^-. If you mix 1.15 g $Na_2C_2O_4$ (FM 134.00) with 0.60 g $KMnO_4$ (FM 158.03) and excess aqueous acid, which reactant is limiting? How much CO_2 is produced?

Limiting Reagent

To decide which reagent is the limiting reagent:

a. Find the number of moles of each available reactant;

b. Compare the number of moles present to the number of moles required for a complete reaction;

c. Determine the limiting one (who is inadequate).

Summary

SI base units: m, kg, s, A, K and mol

Derived quantities: N, Pa, J, L…

Express

Units should be carried along with numbers

Prefixes (kilo-, milli-, micro-) denote multiples of unites

Make solution: $M_{conc} \cdot V_{conc} = M_{dil} \cdot V_{dil}$ with equal mole

Stoichiometry calculation for limiting reagent